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RESEARCH ARTICLE

Genome-wide identification, evolution, and expression analysis of *HVA22* gene family in *Brassica napus* L.

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Abstract The *HVA22* gene family, initially identified in barley, plays a crucial role in plant stress response mechanisms, aiding plants in withstanding various environmental pressures. However, this gene family has not been studied in *Brassica napus* until now. In this study, we identified 39 *HVA22* genes (*BnHVA22.1*–*BnHVA22.39*) distributed across the *B. napus* genome. Phylogenetic analysis categorized these genes into four distinct clades, which include both monocotyledon and dicotyledon species. The *BnHVA22* genes are characterized by a variable number of introns, ranging from three to seven, with intron phases predominantly occurring in phases 1, 0, and 2. Furthermore, the proteins encoded by these genes possess a conserved TB2/DP1/HVA22 domain. The expansion of the *BnHVA22* gene family is primarily attributed to segmental/whole-genome duplications. Duplicated orthologous gene pairs exhibit negative selection, with an average constraint of 0.26, indicating functional conservation of these genes throughout evolution. Stress- and abscisic acid (ABA)-responsive *cis*-regulatory elements are key regulators of *BnHVA22* gene transcription, while microRNAs such as *miRNA167* modulate gene expression at the post-transcriptional level. Transcriptomic analysis has demonstrated the involvement of *BnHVA22* genes in *B. napus* growth, development, and responses to abiotic stress. Moreover, qRT-PCR analysis confirmed the significant differential expression of *BnHVA22.8*, *BnHVA22.14*, *BnHVA22.15*, *BnHVA22.30*, and *BnHVA22.39* in response to ABA and salt stress. These findings elucidate the pivotal role of the *HVA22* gene family in the stress tolerance of *B. napus*, thus providing valuable insights for breeding stress-resistant *B. napus* cultivars.

Keywords Abiotic stress · Abscisic acid · Comparative genomics · Evolution · *HVA22* gene family · Stress responsive gene

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Introduction

Climate change has exposed plants to a wide range of stresses, including drought, salinity, high temperatures, heavy metal contamination, and attacks from pathogens and pests. These stresses significantly impact plant morphology, growth, cellular function, and molecular processes, ultimately restricting plant yield and negatively affecting agricultural productivity (Raza et al. 2019; Fidler et al. 2018; Vishwakarma et al. 2017). As sessile organisms, plants have evolved sophisticated mechanisms to cope with these challenges, thereby enhancing their adaptability to adverse conditions (Singh et al. 2010). Consequently, understanding the mechanisms underlying plant tolerance to stress is crucial for improving crop resilience (Fidler et al. 2018).

Abscisic acid (ABA) is a critical plant hormone that plays multiple roles in mediating responses to both abiotic and biotic stresses. ABA significantly influences calcium signaling (Edel and Kudla 2016) and regulates the expression of various stress-responsive genes, as well as the physiological and metabolic responses of plants, enabling them to better tolerate stress (Ma et al. 2018; Zhang et al. 2021). One key ABA-responsive gene is *HVA22*, first identified during the discovery of ABA-responsive genes in *Hordeum vulgare* (Shen et al. 1993). Various investigations have demonstrated that *HVA22* genes are involved in the response to abiotic stress. In *Arabidopsis thaliana*, five *HVA22*-coding genes have been identified, each exhibiting distinct expression patterns under different abiotic stresses and across various tissues (Chen et al. 2002). The *Solanum lycopersicum SlHVA22i* gene, a homolog of *A. thaliana AtHva22g/j*, exhibits increased expression under cold, heat, drought, and salinity stress (Wai et al. 2022). In *Solanum tuberosum*, the *HVA22* homolog gene plays a critical role in responding to salinity and drought stress and interacts with the protein StPDI1, participating in the sucrose transport pathway (Eggert et al. 2016). The *ZmHVA22* gene in *Zea mays* is significantly suppressed under high salinity, drought, and cold stress but is induced by high temperature, ethylene, and ABA treatments (Chen et al. 2024) over expression of the *HVA22d* gene from *Citrus clementina* in tobacco has been shown to increase drought tolerance and significantly reduce H_2O_2 content under short-term water deficit conditions (Ferreira et al. 2019). The *HLP1* gene from *Oryza sativa*, a homolog of *HVA22*, significantly enhances resistance to blast disease by disrupting endoplasmic reticulum homeostasis (Meng et al. 2022). Additionally, *OsHLP1* has been identified as an endoplasmic reticulum-phagy receptor and plays a role in *O. sativa* immune responses (Liang et al. 2023).

Proteins encoded by *HVA22* genes are characterized by the presence of three transmembrane domains and are primarily localized to the cell membrane and plasmodesmata (Guo and David Ho 2008; Fernandez-Calvino et al. 2011). From an evolutionary perspective, *HVA22* homologs have been identified in a variety of eukaryotes, including cereal, *A. thaliana*, humans, and mice, but are absent in prokaryotes (Shen et al. 2001). Approximately 367 *HVA22* homologs have been discovered in eukaryotes, all of which contain the conserved TB2/DP1/HVA22 domain (Guo and David Ho 2008; Sharon and Suvarna 2017; Ferreira et al. 2019). A Comprehensive analysis of the *HVA22* family can provide valuable insights into their functional characteristics. Systematic research of the *HVA22* family has been conducted in several species, including *A. thaliana*, *S. lycopersicum*, *Glycine max*, *C. clementina*, and *C. sinensis*, identifying five, 15, 40, six, and six *HVA22* genes in these species, respectively (Chen et al. 2002; Ferreira et al. 2019; Wai et al. 2022).

Brassica napus (rapeseed) is a relatively young cultivated species characterized by low genetic diversity and a complex amphiploid genome (AACC). This genome resulted from the cross between *B. rapa* (AA genome) and *B. oleracea* (CC genome) (Chalhoub et al. 2014). Over the past few decades, *B. napus* has emerged as a significant global crop, ranking second in global production among oilseed crops, following *G. max*. The oil content of *B. napus* ranges from 30.6 to 48.3% of its dry weight. The oil is rich in important fatty acids, including oleic acid (56.8% to 64.92%), palmitic acid (4.18% to 5.01%), and linoleic acid (17.11% to 20.93%). Additionally, alpha-tocopherol constitutes approximately 13.33% to 40.01% of the total oil content (Matthaus et al. 2016). Advances in genome sequencing and computational methods have enabled scientists to identify genes involved in various biological processes, study their evolutionary and functional relationships, and utilize target genes in breeding programs for diverse purposes. Given the crucial role of *HVA22* genes in plant stress responses, this study aimed to identify the *HVA22* gene family in *B. napus* for the first time. We analyzed their phylogenetic relationships, conserved motifs, gene structure, intron phases, and regulatory elements in the promoter region, as well as miRNAs influencing these genes. Using RNA-seq data, we examined gene expression profiles in various tissues and their responses to biotic and abiotic stresses. Finally, we assessed the expression of selected genes in response to salinity and ABA hormone treatment via qRT-PCR.

Materials and methods

Identification of *BnHVA22* gene family members in the *B. napus* genome

To identify the *BnHVA22* genes, the Hidden Markov Model (HMM) profile of the B2_DP1_HVA22 domain (PF03134) was retrieved from the Pfam database (Mistry et al. 2021). This profile was used to search the *B. napus* proteome (Brana_Dar_V5) with HMMER software. The results were evaluated to confirm the presence of the TB2_DP1_HVA22 domain using the SMART database (Letunic et al. 2021). Genes containing this domain were considered as members of the *HVA22* gene family, and their coding sequences (CDS), proteins, and promoters were retrieved using the biomart tool from Ensembl Plants (Bolser et al. 2016).

Physicochemical properties of BnHVA22 proteins and cellular localization

To determine the physicochemical properties of BnHVA22 proteins, including molecular weight, protein length, GRAVY, aliphatic index, instability index, and theoretical isoelectric point (pI), the ProtParam tool on the ExPasy server was used (Duvaud et al. 2021). Additionally, the DeepLoc server was employed to predict the intracellular localization of these proteins (Thumuluri et al. 2022).

Alignment and phylogenetic analysis of the *BnHVA22* family

For the phylogenetic analysis of the *BnHVA22* gene family, full-length HVA22 proteins from *A. thaliana*, *O. sativa*, *B. napus*, *S. lycopersicum*, and *H. vulgare* were aligned using the ClustalW server (Larkin et al. 2007). The alignment results were formatted in FASTA format for phylogenetic tree construction using MEGA11 software (Tamura et al. 2021). The neighbor-joining (NJ) algorithm was used to construct the phylogenetic tree, and its reliability was assessed through a bootstrap test with 1000 replicates. The visualization of the tree was conducted using the iTOL server (Letunic and Bork 2024).

Analysis of exon–intron structure, conserved motifs, and functional domains of *BnHVA22* genes

To evaluate the exon–intron structure of *BnHVA22* genes, the GFF3 file of the *B. napus* genome was provided to TBtools software, which generated information on gene structure and intron phases (Chen et al. 2023). Conserved motifs of the proteins encoded by *BnHVA22* genes were identified using the MEME server, while functional domains were identified using the SMART server (Bailey et al. 2009; Letunic et al. 2021).

Chromosomal location, duplication, and collinearity

The positions of *BnHVA22* genes on the chromosomes were extracted from the *B. napus* GFF3 file. Duplication types and paralogous gene pairs of *BnHVA22* were identified using the Multiple Collinearity Scan toolkit X (MCScanX) (Wang et al. 2012). Chromosomal locations of *BnHVA22* genes and relationships between paralogous genes were visualized using the Gene Location Visualize (advanced) tool of TBtools-II (Chen et al. 2023). To investigate the selection pressure on duplicated *BnHVA22* genes, nonsynonymous (Ka), synonymous (Ks) substitutions, and Ka/Ks values were calculated using the simple Ka/Ks calculator tool of TBtools-II (Chen et al. 2023). Ka/Ks ratios of less than one, equal to one, and greater than one were interpreted as indicators of purifying, neutral, and positive selection, respectively (Hajiahmadi et al. 2020) Additionally, syntenic relationships between *BnHVA22* genes and their orthologs in *B. rapa*, *B. oleracea*, and *A. thaliana* were identified using MCScanX and visualized using the Dual Synteny Plot tool of TBtools (Chen et al. 2023).

Analysis of *BnHVA22* gene promoter regions

Cis-regulatory elements in the promoters of *BnHVA22* genes were identified using the PlantCare server (Lescot et al. 2002). For this analysis, 1500 base pairs upstream of the start codon (ATG) of the genes were retrieved from the Ensembl Plants database and submitted in FASTA format to the PlantCare server.

Post-transcriptional regulation of *BnHVA22* genes

To explore potential post-transcriptional regulation of *BnHVA22* genes expression, we used the psRNAtarget server to identify microRNAs (miRNAs) that may target these genes (Dai et al. 2018). The results were visualized using the SRplot server (Tang et al. 2023).

Expression pattern of BnHVA22 based on transcriptomic data

RNA sequencing (RNA-seq) data was used to study the expression patterns of *BnHVA22* genes in different tissues and under stress conditions. We analyzed the response of *BnHVA22* genes to salt, cold, heat, polyethylene glycol (PEG)-induced drought, and ABA treatment using information from the BrassicaEDB database (Chao et al. 2020). RNA-seq data with project ID CRA001775, available at the National Genomics Data Center (NGDC) (https://ngdc.cncb.ac.cn/) was used to examine the expression patterns of *BnHVA22* genes in various tissues, including root, stem, flower, silique, leaf, and seed. *BnHVA22* Genes that had a TPM > 2 in at least one tissue or stress condition were considered transcriptionally active (Wagner et al. 2013). Expression values for each gene were converted to Log2 (TPM + 1) and visualized as a heatmap using TBtools (Chen et al. 2023).

Plant material and treatment

In this study, seeds of the *B. napus* ZS11 cultivar were sterilized with 95% ethanol and placed on moist filter paper to germinate. After five days, the seedlings were transferred to pots containing vermiculite and placed in a growth chamber set at 22 °C with 60% relative humidity and a 14/10 light/dark cycle. The seedlings were watered every other day with half-strength Hoagland's nutrient solution. When the seedlings were four weeks old, they were subjected to salt stress and ABA treatments. For the salt treatment, the seedlings were irrigated with half-strength Hoagland's nutrient solution containing 200 mM NaCl. For ABA treatment, the hormone was sprayed on the leaves at a concentration of 100 μM. Sampling was performed at 0, 3, 6, 12, and 24 h post-treatment. The samples were immediately frozen in liquid nitrogen and stored at -80 °C.

RNA extraction, cDNA synthesis, and qRT-PCR

To study the response of the candidate *BnHVA22* genes to salt stress and ABA, total RNA was extracted from *B. napus* leaf tissue using TRIzol reagent (Vazyme, Nanjing, China) following the manufacturer's instructions. The purity and concentration of The extracted RNA were assessed using a NanoDrop 2000 spectrophotometer (Thermo Scientific, Waltham, MA, USA), and its integrity was evaluated by agarose gel electrophoresis. The extracted RNA was then treated with DNase I to remove genomic DNA contamination. cDNA synthesis was performed using 1 μg of RNA, following the HiScript II Q RT SuperMix for qPCR instructions (Vazyme, Nanjing, China). The qRT-PCR reaction was conducted using TB Green Advantage qPCR premix (TaKaRa, Shiga, Japan) in a CFX384 Real-Time PCR detection system (Bio-Rad, Hercules, CA, USA). Primers for *Actin 7* as an internal control and the candidate *BnHVA22* genes were designed using the online Primer3 server (Supplementary Table 1). The qRT-PCR program was set as follows: 95 °C for 10 min, followed by 40 cycles of 95 °C for 20 s, 60 °C for 20 s, and 75 °C for 20 s. This experiment was conducted in three biological replicates, each with three technical repeats. data analysis was performed using the $2^{-\Delta\Delta Ct}$ method (Livak and Schmittgen 2001). Statistical differences determined with the Student's t-test (*: $p < 0.05$; **: $p < 0.01$; ***: $p < 0.001$).

Results

The *HVA22* gene family in *B. napus* and their physicochemical properties

Using the HMM profile of the HVA22 domain (PF03134), 39 genes encoding HVA22 proteins were identified in the *B. napus* genome and designated *BnHVA22.1* to *BnHVA22.39* based on their chromosomal locations (Table 1). Physicochemical analysis of the encoded proteins revealed an average length of 282/199 amino acids, an average molecular weight of 22.735 kDa, and an average pI of 8.301. The shortest

Table 1 General characteristics of the *HVA22* gene family in *Brassica napus*

Acc. number	Gene name	Chromosome	Start	End	Length	Weight	pI	GRAVY	Aliphatic index	Instability index	Localization
BnaA01g01110D	*BnHVA22.1*	BnA01	596,450	597,906	210	23.4	7.57	−0.07	96.99	50.14	Endoplasmic reticulum
BnaA01g14320D	*BnHVA22.2*	BnA01	7,236,172	7,237,381	116	13.22	10.27	0.25	99.13	30.16	Endoplasmic reticulum
BnaA02g14490D	*BnHVA22.3*	BnA02	8,200,831	8,202,439	185	21.86	6.27	−0.22	91.14	45.33	Endoplasmic reticulum
BnaA02g17400D	*BnHVA22.4*	BnA02	10,523,543	10,528,189	222	25.97	9.61	−0.32	79.46	36.75	Endoplasmic reticulum
BnaA02g33420D	*BnHVA22.5*	BnA02	23,979,715	23,981,193	165	18.45	5.73	0.18	102.93	31.74	Endoplasmic reticulum
BnaA03g13570D	*BnHVA22.6*	BnA03	6,193,660	6,195,304	124	14.49	9.55	0.62	127.64	35.03	Endoplasmic reticulum
BnaA03g16500D	*BnHVA22.7*	BnA03	7,712,442	7,714,700	229	26.25	9.75	−0.18	84.3	46.55	Endoplasmic reticulum
BnaA03g47030D	*BnHVA22.8*	BnA03	24,122,335	24,123,580	136	15.76	9.34	0.2	108.3	27.14	Endoplasmic reticulum
BnaA04g24690D	*BnHVA22.9*	BnA04	17,985,316	17,986,245	159	18.51	9.45	0.23	103.67	39.32	Endoplasmic reticulum
BnaA05g03010D	*BnHVA22.10*	BnA05	1,627,737	1,628,786	159	18.56	8.87	0.17	99.94	39.93	Endoplasmic reticulum
BnaA05g08230D	*BnHVA22.11*	BnA05	4,541,478	4,543,465	256	29.24	9.44	−0.27	80.39	51.94	Endoplasmic reticulum
BnaA06g14160D	*BnHVA22.12*	BnA06	7,645,851	7,648,016	314	35.19	10.01	−0.56	69.23	58.24	Endoplasmic reticulum
BnaA06g21930D	*BnHVA22.13*	BnA06	15,360,312	15,361,415	206	23.21	6.26	0.29	107.95	42.57	Endoplasmic reticulum
BnaA06g37100D	*BnHVA22.14*	BnA06	24,193,905	24,195,816	289	32.31	6.55	−0.59	64.65	53.82	Endoplasmic reticulum
BnaA07g31630D	*BnHVA22.15*	BnA07	22,045,667	22,046,871	180	20.98	8.35	−0.08	94.19	37.98	Endoplasmic reticulum
BnaA08g15390D	*BnHVA22.16*	BnA08	12,807,082	12,809,151	208	23.24	5.61	−0.05	94.15	55.42	Endoplasmic reticulum
BnaA08g31230D	*BnHVA22.17*	BnA08r	1,894,354	1,896,672	311	34.86	10.19	−0.55	71.81	55.89	Endoplasmic reticulum
BnaAnng07690D	*BnHVA22.18*	BnAnnr	7,427,277	7,429,367	260	29.5	9.65	−0.26	82.12	48.78	Endoplasmic reticulum
BnaAnng09020D	*BnHVA22.19*	BnAnnr	9,565,617	9,566,627	188	21.8	7.49	−0.31	77.7	53.92	Endoplasmic reticulum
BnaAnng21830D	*BnHVA22.20*	BnAnnr	24,348,423	24,349,919	165	18.4	5.63	0.2	103.54	28.67	Endoplasmic reticulum
BnaAnng26710D	*BnHVA22.21*	BnAnnr	30,640,071	30,641,232	140	16.51	6.79	0.22	110	44.07	Endoplasmic reticulum
BnaC01g02140D	*BnHVA22.22*	BnC01	1,099,434	1,100,761	211	23.32	7.57	−0.02	97	51.46	Peroxisome
BnaC01g16810D	*BnHVA22.23*	BnC01	11,478,936	11,480,175	136	15.77	8.85	0.24	99.63	26.72	Endoplasmic reticulum
BnaC02g19470D	*BnHVA22.24*	BnC02	15,856,240	15,857,333	111	12.69	4.63	0.61	130.27	40.55	Endoplasmic reticulum
BnaC03g16470D	*BnHVA22.25*	BnC03	8,371,285	8,372,860	124	14.49	9.55	0.61	127.64	36.6	Endoplasmic reticulum
BnaC03g19880D	*BnHVA22.26*	BnC03	10,409,300	10,410,620	237	26.63	9.43	−0.07	89.28	49.02	Endoplasmic reticulum
BnaC03g51640D	*BnHVA22.27*	BnC03	36,262,927	36,265,718	234	26.53	9.75	0.16	107.21	54.64	Peroxisome
BnaC03g61850D	*BnHVA22.28*	BnC03	51,043,048	51,046,165	207	23.11	6.01	0	96.99	51.85	Endoplasmic reticulum
BnaC04g09250D	*BnHVA22.29*	BnC04	7,005,380	7,007,460	256	29.24	9.62	−0.27	82.31	52.57	Endoplasmic reticulum
BnaC04g17900D	*BnHVA22.30*	BnC04	16,641,409	16,643,520	291	32.96	6.94	−0.61	66.24	52.4	Endoplasmic reticulum
BnaC04g48540D	*BnHVA22.31*	BnC04	47,092,706	47,093,541	159	18.53	9.45	0.21	103.67	38.37	Endoplasmic reticulum
BnaC04g52610D	*BnHVA22.32*	BnC04r	464,348	465,389	159	18.57	8.87	0.2	102.41	39.93	Endoplasmic reticulum

Table 1 (continued)

Acc. number	Gene name	Chromosome	Start	End	Length	Weight	pI	GRAVY	Aliphatic index	Instability index	Localization
BnaC05g15520D	*BnHVA22.33*	BnC05	9,389,295	9,393,032	314	35.33	9.89	−0.55	68.91	58.54	Endoplasmic reticulum
BnaC06g27930D	*BnHVA22.34*	BnC06	29,258,649	29,260,458	187	21.94	6.52	−0.18	89.62	43.7	Endoplasmic reticulum
BnaC06g36360D	*BnHVA22.35*	BnC06	34,946,041	34,947,483	188	21.83	7.56	−0.35	75.61	51.7	Endoplasmic reticulum
BnaC07g39210D	*BnHVA22.36*	BnC07	40,210,953	40,211,998	149	17.2	10.12	0.05	103.38	27.23	Endoplasmic reticulum
BnaC08g19110D	*BnHVA22.37*	BnC08	22,083,695	22,085,986	311	34.86	10.15	−0.53	72.42	58.48	Endoplasmic reticulum
BnaCnng67550D	*BnHVA22.38*	BnCnnr	67,239,668	67,240,940	117	13.27	9.5	−0.05	89.14	26.3	Endoplasmic reticulum
BnaCnng68090D	*BnHVA22.39*	BnCnnr	67,689,126	67,690,027	159	18.69	6.95	−0.15	90.7	37.22	Endoplasmic reticulum

protein, BnHVA22.24, contains 111 amino acids and has a molecular weight of 12.69 kDa, while The longest proteins, BnHVA22.33 and BnHVA22.12, each consist of 314 amino acids. BnHVA22.33 also exhibited the highest molecular weight at 35.33 kDa. The highest pI, 10.3, was observed for BnHVA22.2, whereas the lowest pI, 4.63, was recorded for BnHVA22.24 (Table 1). The average aliphatic index and instability index of the analyzed proteins were found to be 93.37 and 43.83, respectively. Among these proteins, *BnHVA22.14* exhibited the lowest aliphatic index (64.65), whereas *BnHVA22.24* displayed the highest value (130) (Table 1). Regarding protein stability, *BnHVA22.38* showed the lowest instability index (26.3), while *BnHVA22.33* exhibited the highest instability index (58.84). Additionally, the GRAVY (Grand Average of Hydropathy) values for *BnHVA22* proteins averaged 0.046, ranging from -0.61 for *BnHVA22.30* to 0.64 for *BnHVA22.6* (Table 1).

To further investigate the proteins encoded by *BnHVA22* genes, conserved motifs in these proteins were evaluated using the MEME server. Twenty conserved motifs were identified (Fig. 1). The lengths of these motifs ranged from six amino acids in motifs 10 and 17 to 50 amino acids in motifs 4, 7, 9, 12, and 18. The most abundant motifs were motifs 1, 2, and 3, with frequencies of 38, 38, and 37, respectively, all of which are functionally related to the TB2/DP1/HVA22 family (Fig. 1, Supplementary Table 2). Among the identified motifs, motif 1 was found in all BnHVA22 proteins except for BnHVA22.24, motif 2 was found in all BnHVA22 proteins except for BnHVA22.2, and motif 3 was present in all BnHVA22 proteins except for BnHVA22.26 and BnHVA22.27. Motif 18 was found only in BnHVA22.14 and BnHVA22.30, while motif 20 was observed only in BnHVA22.4 and BnHVA22.38 (Fig. 1). Localization analysis of *BnHVA22* proteins showed that, except for BnHVA22.22 and BnHVA22.27, which are active in the peroxisome, all other identified proteins are localized to the endoplasmic reticulum (Table 1).

Phylogenetic analysis of the *HVA22* gene family

To examine the evolutionary relationships within the *BnHVA22* gene family, a phylogenetic tree was constructed using the NJ method. The analysis was based on HVA22 protein sequences from *B. napus*, *O. sativa*, *H. vulgare*, *A. thaliana*, and *S. lycopersicum*.

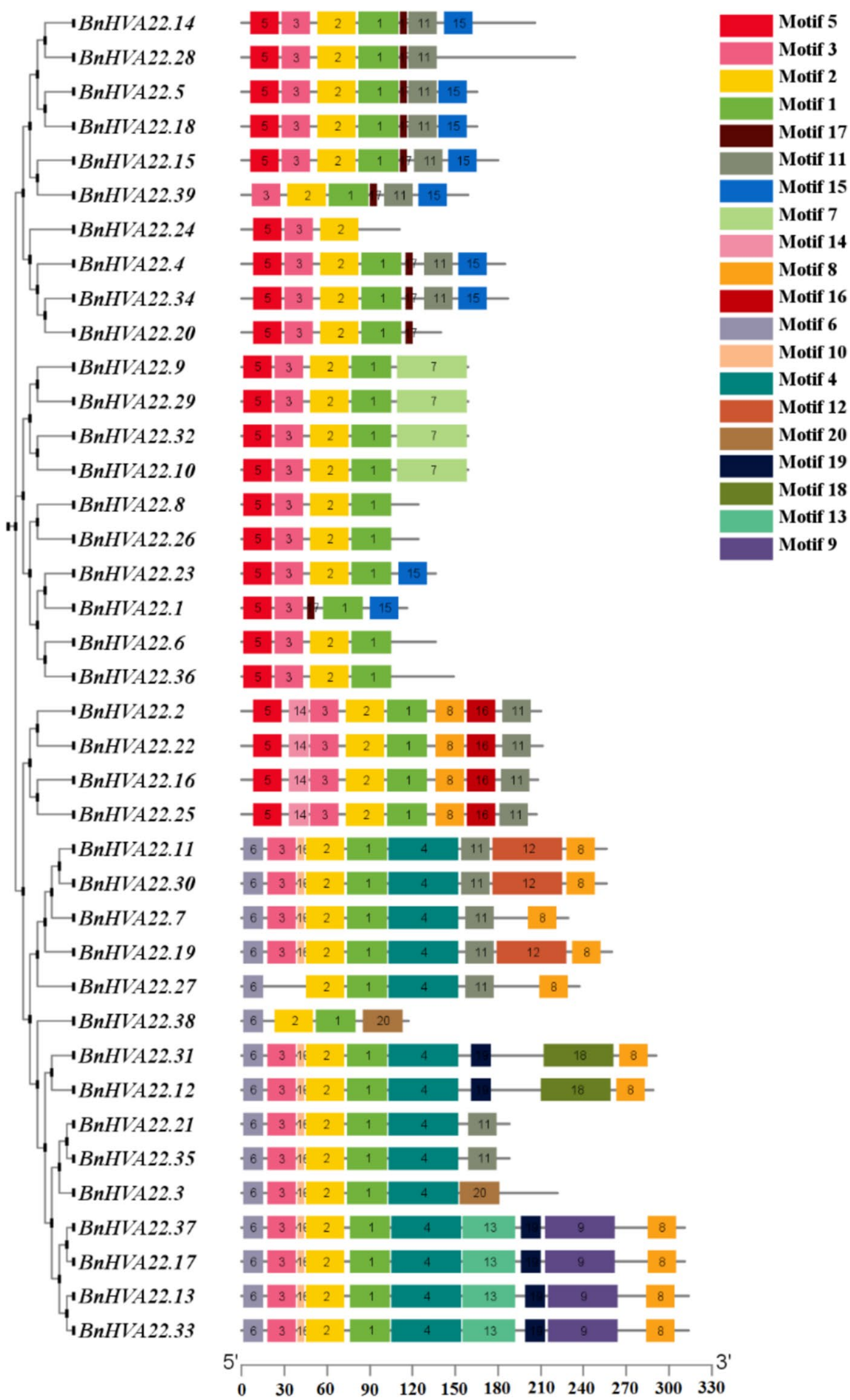


Fig. 1 Identified conserved motifs in *BnHVA22* genes. A total of 20 conserved motifs were identified, represented in various colored boxes. The length of the proteins is indicated based on the scale bar at the bottom

This tree included 11 *AtHVA22* from *A. thaliana*, 13 *HvHVA22* from *H. vulgare*, 14 *SlHVA22* from *S. lycopersicum*, 15 *OsHVA22* from *O. sativa*, and 39 *BnHVA22* from *B. napus*. Phylogenetic analysis revealed a clear division of the *HVA22* genes into four distinct clades (Fig. 2). Clade A contained nine

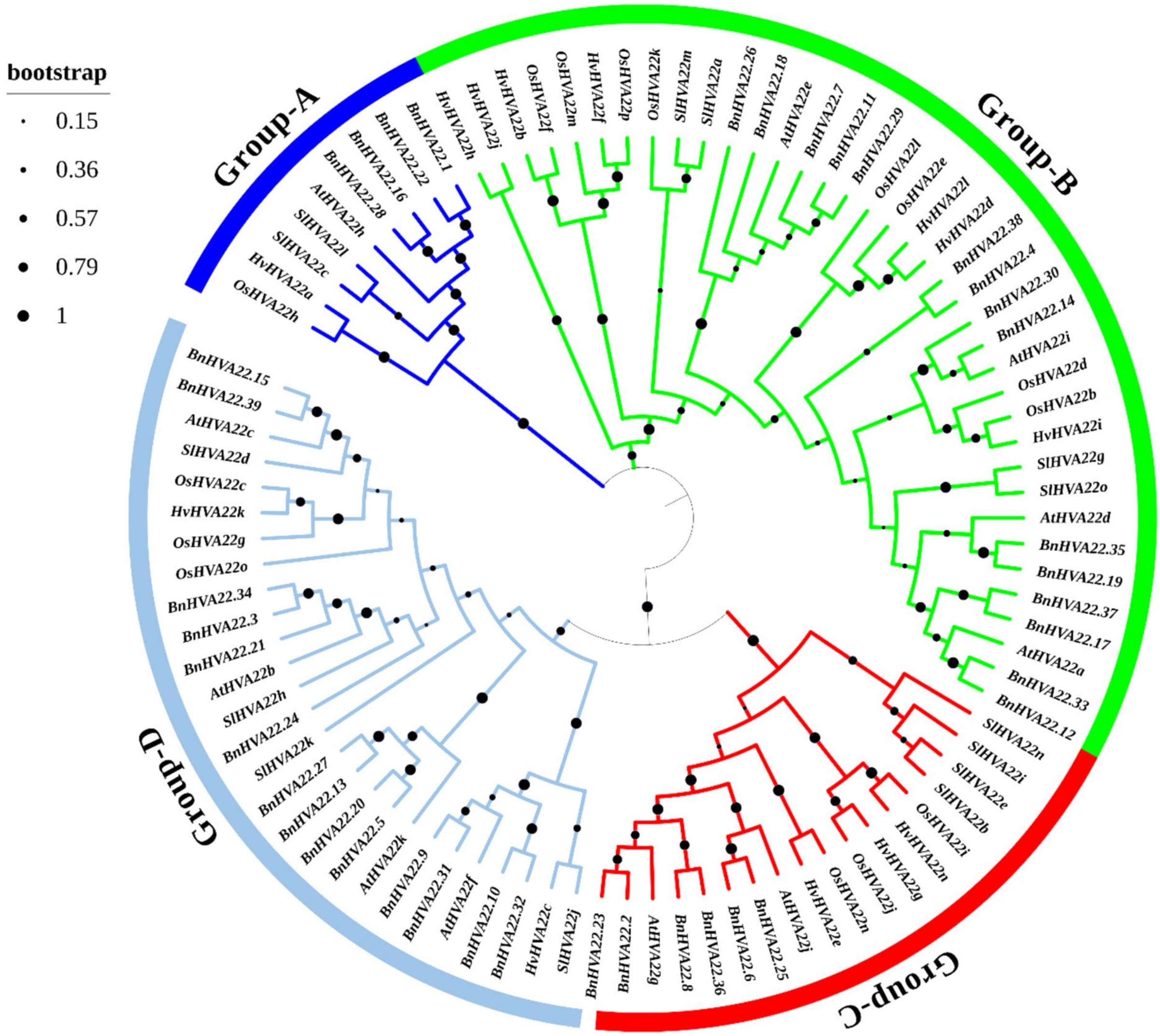


Fig. 2 Phylogenetic relationships of *BnHVA22* genes. The phylogenetic tree was constructed using MEGA11 software and the neighbor-joining method. The tree includes *HVA22* genes from *Brassica napus* (*BnHVA22*), *O. sativa* (*OsHVA22*), *A. thaliana* (*AtHVA22*), *H. vulgare* (*HvHVA22*), and *S. lycopersicum* (*SlHVA22*). Each clade is marked with a distinct color

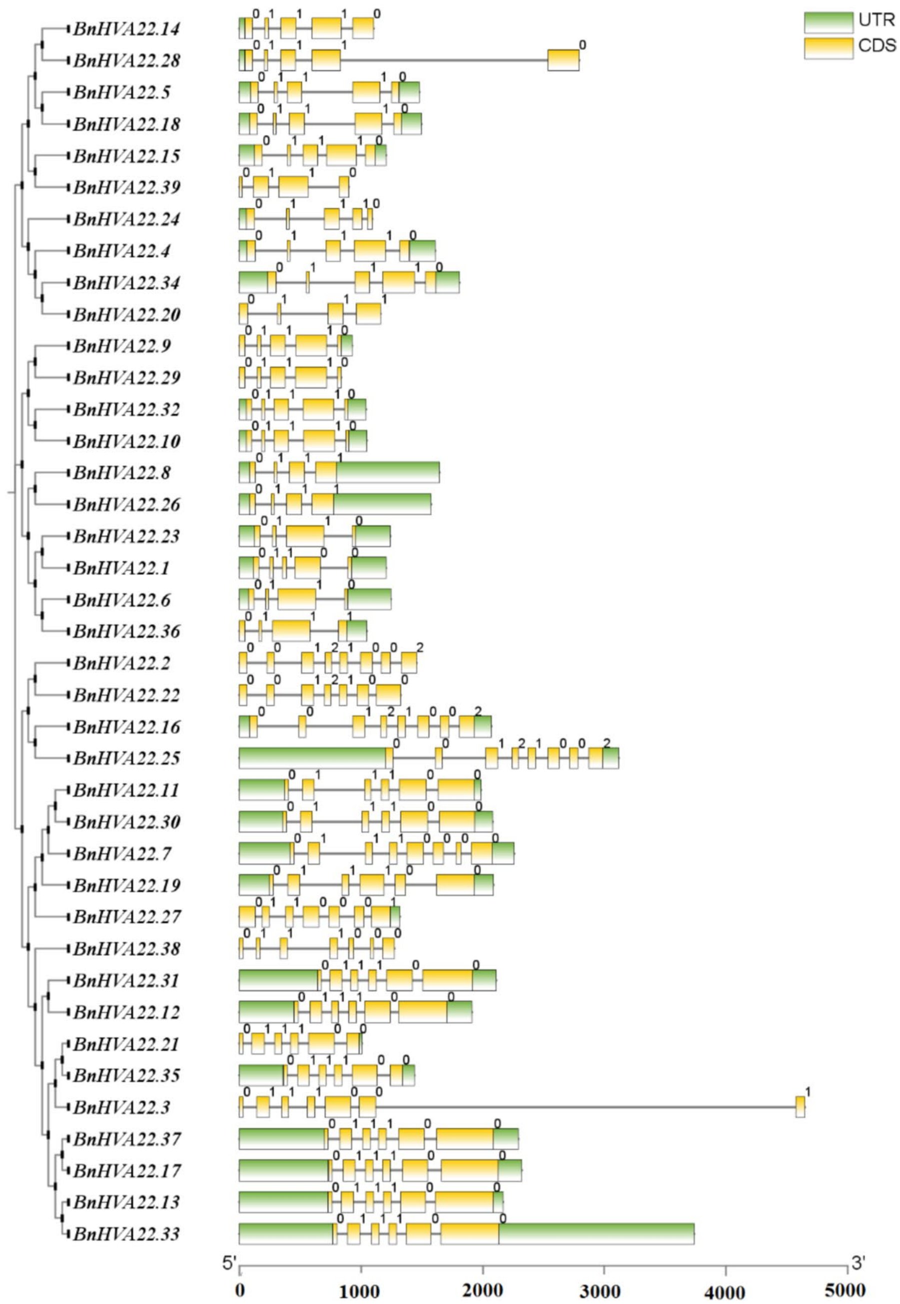
genes (one *AtHVA22*, one *HvHVA22*, two *SlHVA22*, one *OsHVA22*, and four *BnHVA22*). Clade B included 38 genes (four *AtHVA22*, seven *HvHVA22*, four *SlHVA22*, eight *OsHVA22*, and 15 *BnHVA22*). Clade C comprised 18 genes (two *AtHVA22*, three *HvHVA22*, four *SlHVA22*, three *OsHVA22*, and six *BnHVA22*). Finally, clade D contained 27 genes (four *AtHVA22*, two *HvHVA22*, four *SlHVA22*, three *OsHVA22*, and 14 *BnHVA22*) (Fig. 2).

Interestingly, HVA22 genes from *H. vulgare* and *O. sativa* consistently clustered together within each clade. Similarly, *B. napus* and *A. thaliana* HVA22 genes formed distinct clusters (Fig. 2). This pattern likely reflects the evolutionary relationships between these plant species. *H. vulgare* and *O. sativa* belong to the Poaceae family, while *B. napus* and *A. thaliana* are members of the Brassicaceae family. In contrast, *S. lycopersicum*, a dicotyledonous plant, exhibited a closer evolutionary relationship with the dicotyledonous *B. napus* and *A. thaliana* than with the monocotyledonous *H. vulgare* and *O. sativa*.

Exon–intron structure and intron phase of *BnHVA22* genes

Analysis of the exon–intron structure of the *BnHVA22* gene family revealed a range of three to seven introns. The lowest number of introns was found in *BnHVA22.6*, *BnHVA22.21*, *BnHVA22.25*, *BnHVA22.36*, *BnHVA22.8*, *BnHVA22.23*, and *BnHVA22.39* (Fig. 3). In contrast, the highest number of introns was observed in *BnHVA22.1*, *BnHVA22.16*, *BnHVA22.28*, and *BnHVA22.7*. Additionally, 13 genes had four introns, 11 genes had five introns, and four genes had six introns. *BnHVA22.36* was identified as the longest gene in the family due to its extended sixth intron, longer untranslated regions (UTRs), and a greater overall number of introns (Fig. 3).

Fig. 3 Exon–intron structure of *BnHVA22* gene family members. Exons, introns, and UTR regions are represented by yellow boxes, black lines, and green boxes, respectively. The length of the genes is indicated based on the scale bar at the bottom



Intron phase analysis revealed that the *BnHVA22* genes could be divided into 10 different groups: 0110, 0111, 01110, 011100, 0111001, 0110001, 01100010, 0012100, 00121002, and 01110000 (Fig. 3). Phase zero and one introns were present in all *BnHVA22* genes. However, only *BnHVA22.1*, *BnHVA22.16*, *BnHVA22.22*, and *BnHVA22.28* had phase two introns, in addition to phase zero and one. The percentages of introns with phase zero, phase one, and phase two were calculated as 46.57%, 50.22%, and 3.19%, respectively.

Gene location, duplication, and collinearity

Mapping of *BnHVA22* genes on the *B. napus* genome revealed that 16 genes are located on the A sub-genome chromosomes, and 15 are located on the C sub-genome chromosomes (Fig. 4). Additionally, eight genes were found on the scaffolds A08_random, Ann_random, C0_random, and Cnn_random. Chromosome C03 harbored The highest number of genes, four, while chromosomes A04, A07, A08, C02, C05, C07, and C08 each had only one gene (Fig. 4).

The study of duplication mechanisms affecting the *BnHVA22* gene family, using the MCScanX tool, indicated that *BnAHVA22.20*, *BnAHVA22.38*, and *BnAHVA22.39* arose from dispersed duplication, while the other genes were created by segmental/whole genome duplication (WGD). These findings suggest that segmental/WGD duplication may play a significant role in the expansion of *HVA22* genes in *B. napus* (Fig. 4). Intraspecific collinearity analysis revealed 27 pairs of *BnHVA22* paralogous genes in the *B. napus* genome (Fig. 4). To investigate the selection pressures on paralogous genes, Ka, Ks, and Ka/Ks values were calculated for each paralogous gene pair. The analysis indicated that duplicated gene pairs have been under purifying selection pressure during evolution, with an average Ka/Ks ratio of 0.26 (Supplementary Table 3).

Furthermore, interspecific collinearity analysis between the *B. napus* genome and those of *B. rapa*, *B. oleracea*, and *A. thaliana* revealed 23 pairs of orthologous genes between 18 *BnHVA22* genes and 11 *AtHVA22* genes (Fig. 5a), 44 pairs between 36 *BnHVA22* genes and 18 *BoHVA22* genes (Fig. 5b), and 43 pairs between 36 *BnHVA22* genes and 18 *BrHVA22* genes (Fig. 5c).

Promoter analysis of *BnHVA22* genes and identification of effective miRNA molecules

Cis-regulatory elements in the promoter region of genes are essential for regulating gene expression under various biological conditions. To understand the regulatory mechanisms of *BnHVA22* genes, their promoters were analyzed using the PlantCARE server. In silico analysis identified four major groups of *cis*-regulatory elements: those responding to tissue growth and development, hormones, promoter-dependent regulators, and stress (Supplementary Table 4).

Specifically, 19 types of tissue growth and development regulatory elements were found, including those involved in endosperm growth, secondary xylem development, meristem activity, cambium formation, cell cycle regulation, circadian rhythm, germination, flavonoid biosynthesis, zein metabolism, alpha-amylase activity, and receptor signaling (Supplementary Table 4). Twelve types of hormone-related regulatory elements were identified, responding to auxin, gibberellin, ethylene, ABA, and salicylic acid hormones (Supplementary Table 4). Five types of regulatory elements associated with promoter DNA, AT-rich regions, binding proteins, MYBHv1, and other regulatory elements were also detected. Additionally, 17 types of regulatory elements related to stress conditions, such as cadmium, anaerobic conditions, wounds, anoxia, low temperature, and drought were identified (Supplementary Table 4). The most abundant *cis*-regulatory elements in the tissue growth and development group were the AAGAA-motif and MYb-binding site. In the plant hormone group, ABRE and ERE elements were the most prevalent. For protein-binding sites, the CCAAT-box was most common. Finally, in the stress group, MYc and MYb elements were the most abundant (Supplementary Table 4).

MicroRNAs (miRNAs) are another important factor in regulating gene expression. In the case of *BnHVA22* genes, 19 miRNAs from five different families were found to target 10 of these genes (Fig. 6). The bna-miR6028 miRNA, which targets four genes (*BnHVA22.3*, *BnHVA22.21*, *BnHVA22.24*, and *BnHVA22.34*), had the most significant impact. In contrast, bna-miR6032 and bna-miR167a-d had the least impact, targeting *BnHVA22.37* and *BnHVA22.33*, respectively. Other miRNAs, such as bna-miR169 and

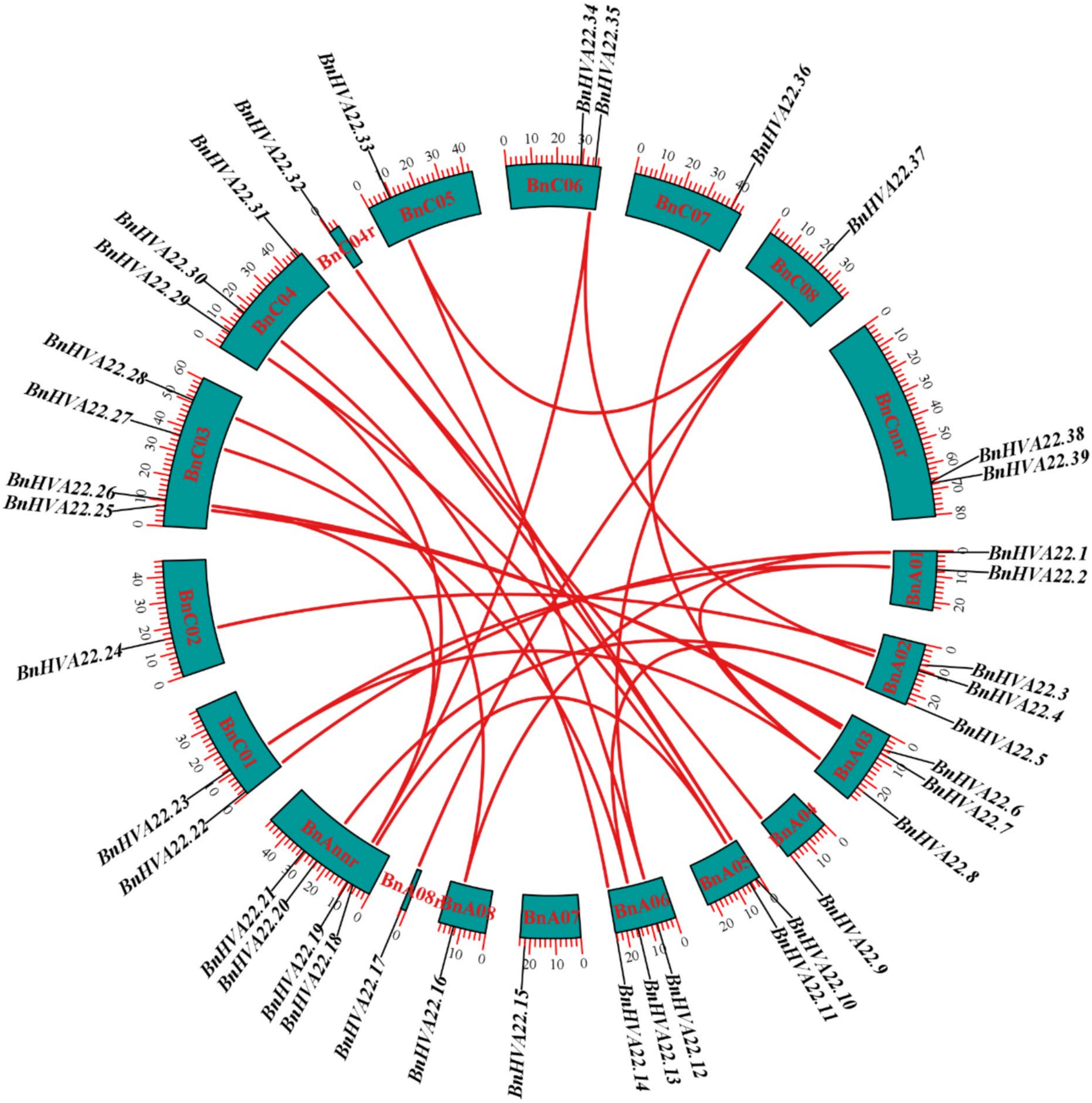


Fig. 4 Genomic locations of *BnHVA22* genes on *Brassica napus* chromosomes and relationships between paralogous genes. Chromosomes are colored blue, and the relationships between paralogous pairs genes are shown with red curves

bna-miR172a-d, targeted *BnHVA22.2/BnHVA22.16* and *BnHVA22.14/BnHVA22.30*, respectively (Fig. 6).

Expression profile of *BnHVA22* genes based on transcriptomic data

Gene expression profiling of the *BnHVA22* family, based on RNA-seq data, revealed that 27 genes from this family were expressed (TPM $\geq$ 2) in at least one of the root, stem, leaf, flower, silique, or seed tissues, or under stress conditions such as salt, PEG, cold, heat, and ABA (Fig. 7). Based on the expression patterns, these genes were categorized into three groups.

Group one included *BnHVA22.14, BnHVA22.15, BnHVA22.30*, and *BnHVA22.39*, which were expressed (TPM > 2) in response to all stress

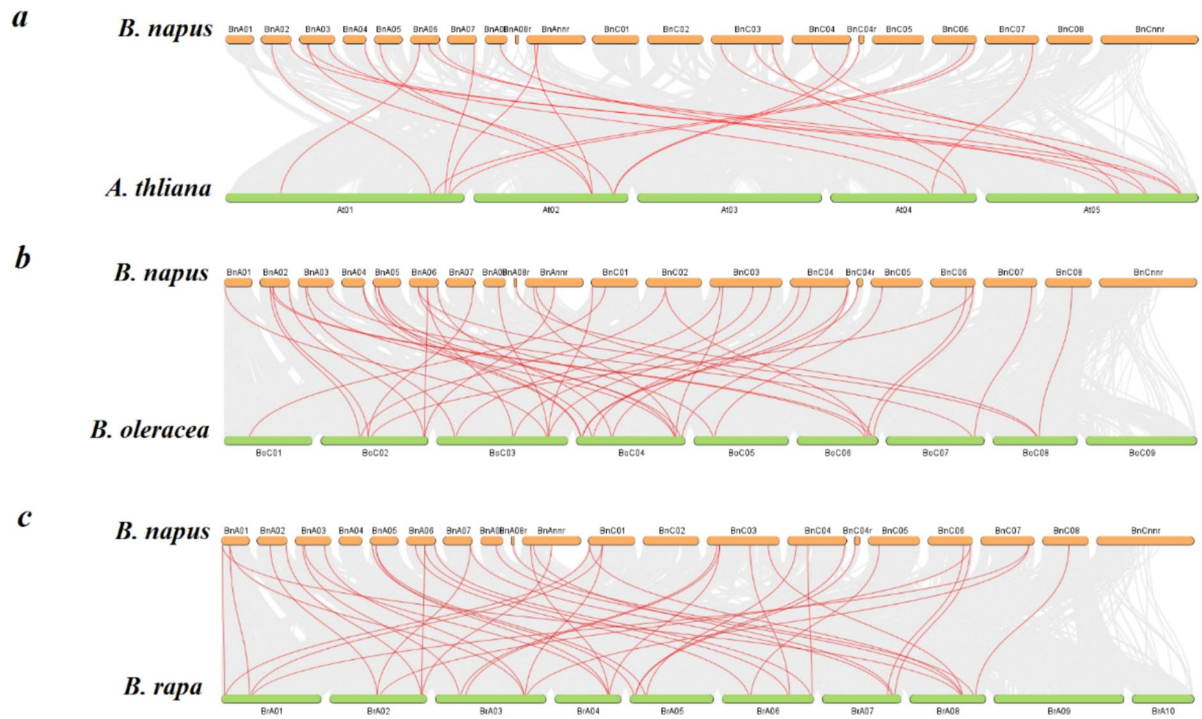


Fig. 5 Collinearity analysis of *HVA22* genes between *Brassica napus*, *Arabidopsis thaliana*, *B. rapa*, and *B. oleracea*. Gray and red lines represent collinear blocks and pairs of orthologous genes among the studied species, respectively

conditions and in all tissues. The second group consisted of *BnHVA22.2, BnHVA22.8, BnHVA22.23*, and *BnHVA22.36* (Fig. 7). With the exception of *BnHVA22.8*, the genes in this group were expressed in all tissues but displayed varied expression patterns in response to stress. For example, *BnHVA22.2* was exclusively expressed in response to cold stress, while *BnHVA22.8* was not expressed under PEG stress.

Group three was tissue-specific, and with the exception of *BnHVA22.16*, the other genes in this group did not show significant expression in response to stress (Fig. 7). A detailed examination of the expression pattern of genes in group three revealed that *BnHVA22.7, BnHVA22.11, BnHVA22.18, BnHVA22.26*, and *BnHVA22.29* were specific to flower and leaf tissues. Additionally, *BnHVA22.12, BnHVA22.17, BnHVA22.21, BnHVA22.33, BnHVA22.34*, and *BnHVA22.37* had higher expression in root, seed, and silique tissues. Furthermore, *BnHVA22.6* was expressed in the root, *BnHVA22.10* in the silique, *BnHVA22.9* in both the silique and root, *BnHVA22.3* and *BnHVA22.24* in the seed, and *BnHVA22.16* in the flower and in response to heat stress (Fig. 7). As observed, the expression patterns of individual *BnHVA22* genes are distinct, yet there is an overlap in their expression across different tissues and in response to stress. These findings suggest that each *BnHVA22* gene has a specific functional role, while also cooperating in various biological processes in *B. napus*.

Response of *BnHVA22* genes to salt and ABA

A qRT-PCR analysis of the expression of *BnHVA22.8, BnHVA22.14, BnHVA22.15, BnHVA22.30*, and *BnHVA22.39* in response to salt stress and ABA treatment revealed that, with the exception of *BnHVA22.30*, which showed no significant response to salt stress, the other genes exhibited significant changes in expression at least at one time point under both salt and ABA treatments. *BnHVA22.8* and *BnHVA22.39* exhibited a positive response to salt stress (Fig. 8). Specifically, the expression of *BnHVA22.8* was up-regulated by 3.53-, 3.33-, and 2.13-fold at 6, 12, and 24 h, respectively.

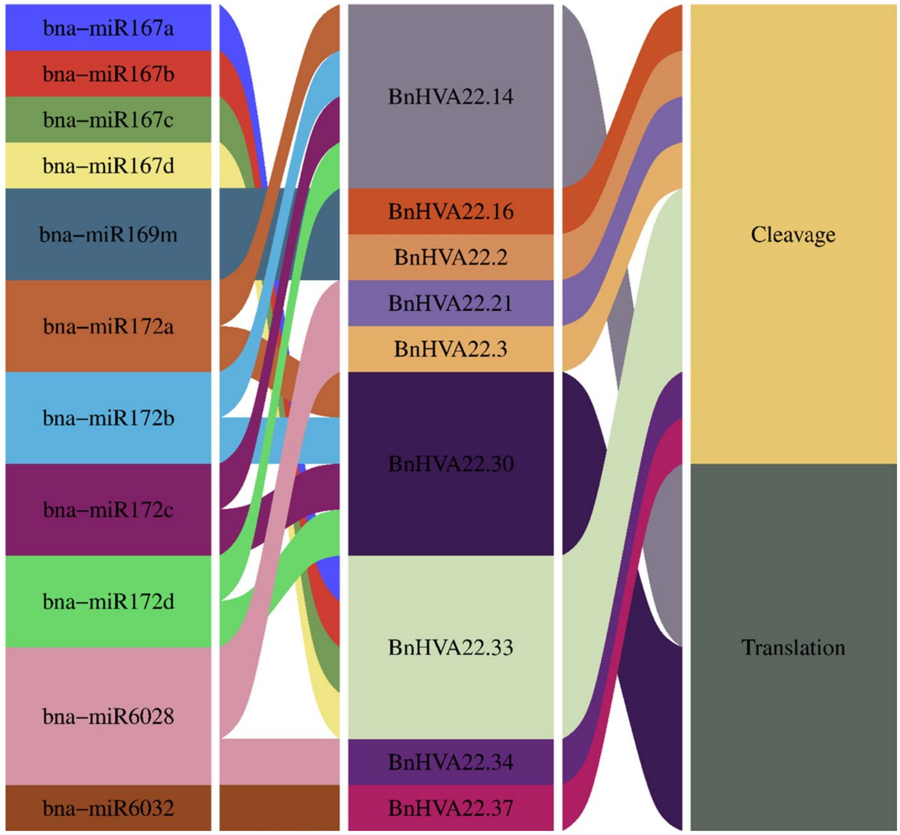


Fig. 6 Relationship between *BnHVA22* genes and miRNA molecules. The columns represent miRNA molecules, *HVA22* genes from *Brassica napus*, and the method of inhibition

The expression of *BnHVA22.39* was up-regulated by 2.09- to 2.7-fold at all-time points examined (Fig. 8).

The expression patterns of *BnHVA22.14* and *BnHVA22.15* were more complex. The expression of *BnHVA22.14* decreased at 3, 6, and 12 h, followed by a significant increase at later time points. At 12 and 24 h post-stress, its expression was suppressed by 7.14-fold and subsequently up-regulated by 2.96-fold, respectively. Similarly, *BnHVA22.15* showed both up-regulation and down-regulation, but only at 6 and 12 h. Specifically, It was suppressed by twofold at 6 h and up-regulated by 2.94-fold at 12 h (Fig. 8).

In response to ABA treatment, *BnHVA22.8, BnHVA22.14*, and *BnHVA22.39* were significantly up-regulated at all-time points (Fig. 9). The expression of *BnHVA22.8* increased fourfold at 3 h, *BnHVA22.14* increased 2.45-fold at 6 h, and *BnHVA22.39* increased 5.22-fold at 12 h. The expression of *BnHVA22.15* also increased in response to ABA, but only significantly at 12 and 24 h, with 3.49-fold and 1.71-fold increases, respectively. In contrast, *BnHVA22.30* showed a more variable response to ABA, with significant up-regulation at 3 h, a decrease at 6 and 12 h, and a subsequent increase at 24 h. (Fig. 9).

Discussion

Plant hormones, such as ABA, act as signaling molecules that regulate various aspects of growth, development, and stress responses (Waadt et al. 2022; Rehman et al. 2021). ABA is particularly important in plant responses to abiotic stress and is often referred to as the "stress hormone(Sah et al. 2016). ABA regulates stress responses by modulating the expression of a broad range of stress-responsive genes. The *HVA22* gene family plays a crucial role in ABA-mediated stress responses. These genes have been identified in various eukaryotes but are absent in prokaryotes (Chen et al. 2024).

Previous research has shown that the number of *HVA22* genes can vary among plant species. For instance, 11, 15, and 40 *HVA22* genes have been identified in *H. vulgare*, *S. lycopersicum*, and *G.*

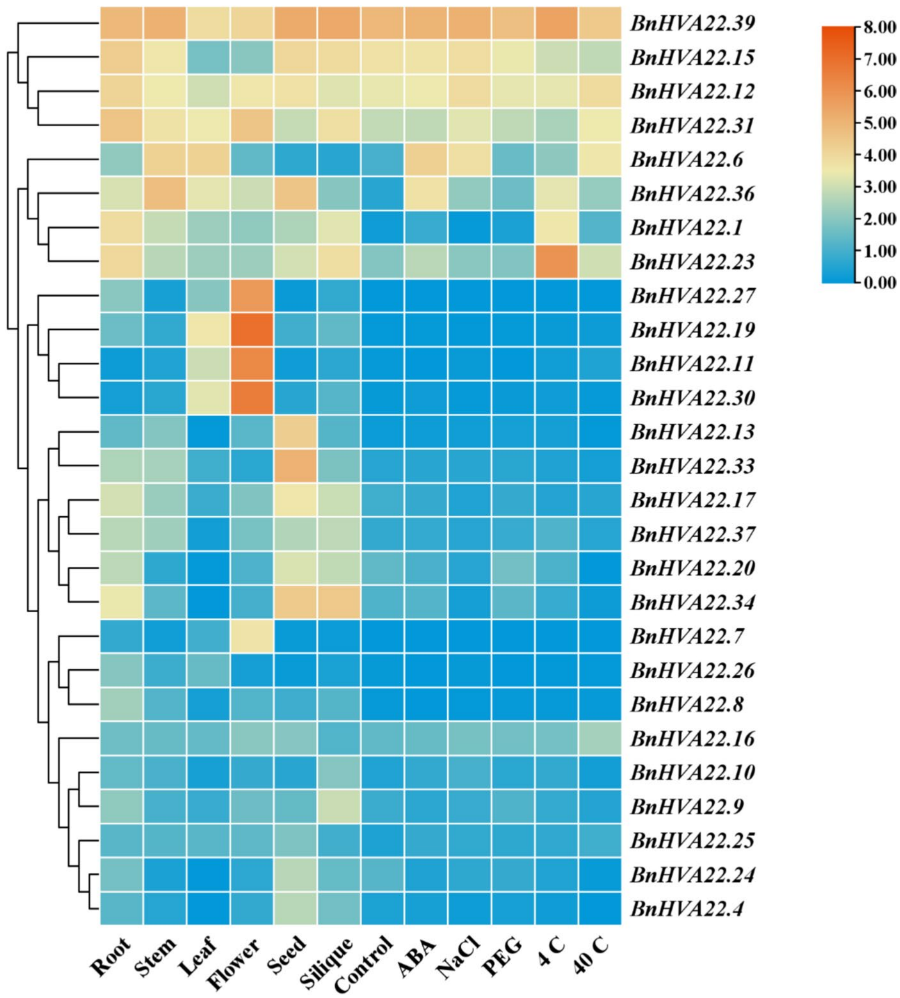


Fig. 7 Expression profile of *BnHVA22* genes in different tissues of *Brassica naups* and response to abiotic stresses. Expression values for the genes were transformed to log2 (TPM + 1) and visualized using a heatmap generated with TBtools

max, respectively (Chen et al. 2002, 2024; Shen et al. 2001; Wai et al. 2022). In different cotton species, *Gossypium barbadense*, *G. hirsutum*, *G. arboreum*, and *G. raimondii* possess 34, 32, 16, and 17 *HVA22* genes, respectively (Zhang et al. 2023a) Additionally, this gene family comprises six members in *C. clementina*, *C. sinensis*, and *Passiflora edulis* (Ferreira et al. 2019; Hou et al. 2024). The variation in the size of the *HVA22* gene family among different plants can be attributed to several factors. While genome size can vary significantly among species (e.g., *S. lycopersicum*: 950 Mbp, *H. vulgare*: 5.1 Gbp, *G. max*: 1.1 Gbp), there is no clear correlation between genome size and the number of *HVA22* genes. However, previous research suggests that the evolutionary history of plants, particularly the occurrence of various duplication events, is a key factor driving the expansion of the *HVA22* gene family in different species.

Phylogenetic analysis revealed that *HVA22* genes are divided into four clades, consistent with previous research (Wai et al. 2022; Hou et al. 2024; Zhao et al. 2023). Each clade contains at least one *HVA22* gene from each of the studied plants, indicating that these genes evolved before the divergence of monocotyledon and dicotyledon species (Wolfe et al. 1989; Wai et al. 2022). The clustering of *HVA22* genes based on their evolutionary relationships highlights the conserved nature of these genes across plant species. The closer evolutionary affinity between *A. thaliana* and *B. napus* compared to *S. lycopersicum* underscores the importance of family-level relationships in shaping gene evolution and function. Genes with similar exon–intron structures and motif distributions

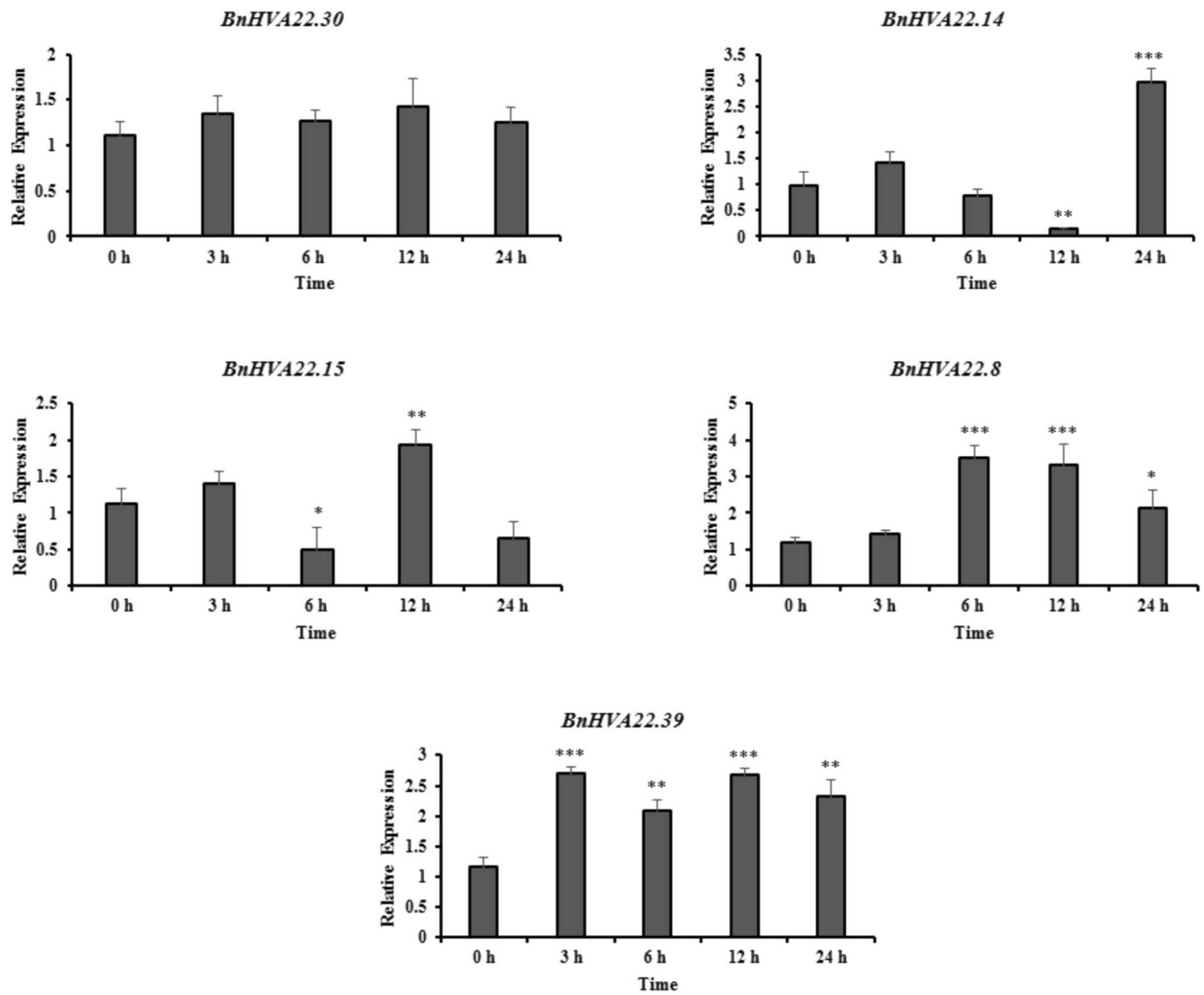


Fig. 8 qRT-PCR analysis of selected *BnHVA22* genes in response to salt stress (200 mM NaCl). Actin 7 was used as internal control. The asterisks indicate significant differences according to Student's t test (*$P<0.05$, **$P<0.01$, *** $P<0.001$)

are likely to have similar functional roles (Sarcheshmeh et al. 2023; Zafar et al. 2022). Our results showed that motifs 1, 2, and 3, with frequencies of 38, 38, and 37, respectively, are related to the TB2/ DP1/ HVA22 family. Except for *BnHVA22.2* and *BnHVA22.24* (which lack domains 2 and 1, respectively), and *BnHVA22.26* and *BnHVA22.27* (which lack domain 3), all three domains were identified in the remaining family members. However, each clade exhibits a unique motif distribution and frequency pattern. For example, motif 16 is specific to the A clade, and motifs 9, 10, and 13 are unique to clade 13. This diversity suggests that genes from different clades may have specialized functions. Research on conserved motifs in HVA22 genes from other plants has shown that these genes contain additional functional domains beyond the TB2/DP1/HVA22 domain. For instance, RVT_motif and ZnF domains have been identified in some *G. max* HVA22 genes, and certain *S. lycopersicum* HVA22 genes also contain the RVT_motif. These domains contribute to the functional diversity of HVA22 genes in these plants (Chen et al. 2024; Wai et al. 2022).

exon–intron structure and intron phase play significant roles in gene evolution (Abedi et al. 2021). Introns are classified into three phases: phase 0, phase 1, and phase 2. Phase 0 introns are inserted between codons, phase 1 introns are located between the first and second nucleotides of a codon, and phase 2 introns occur between the second and third

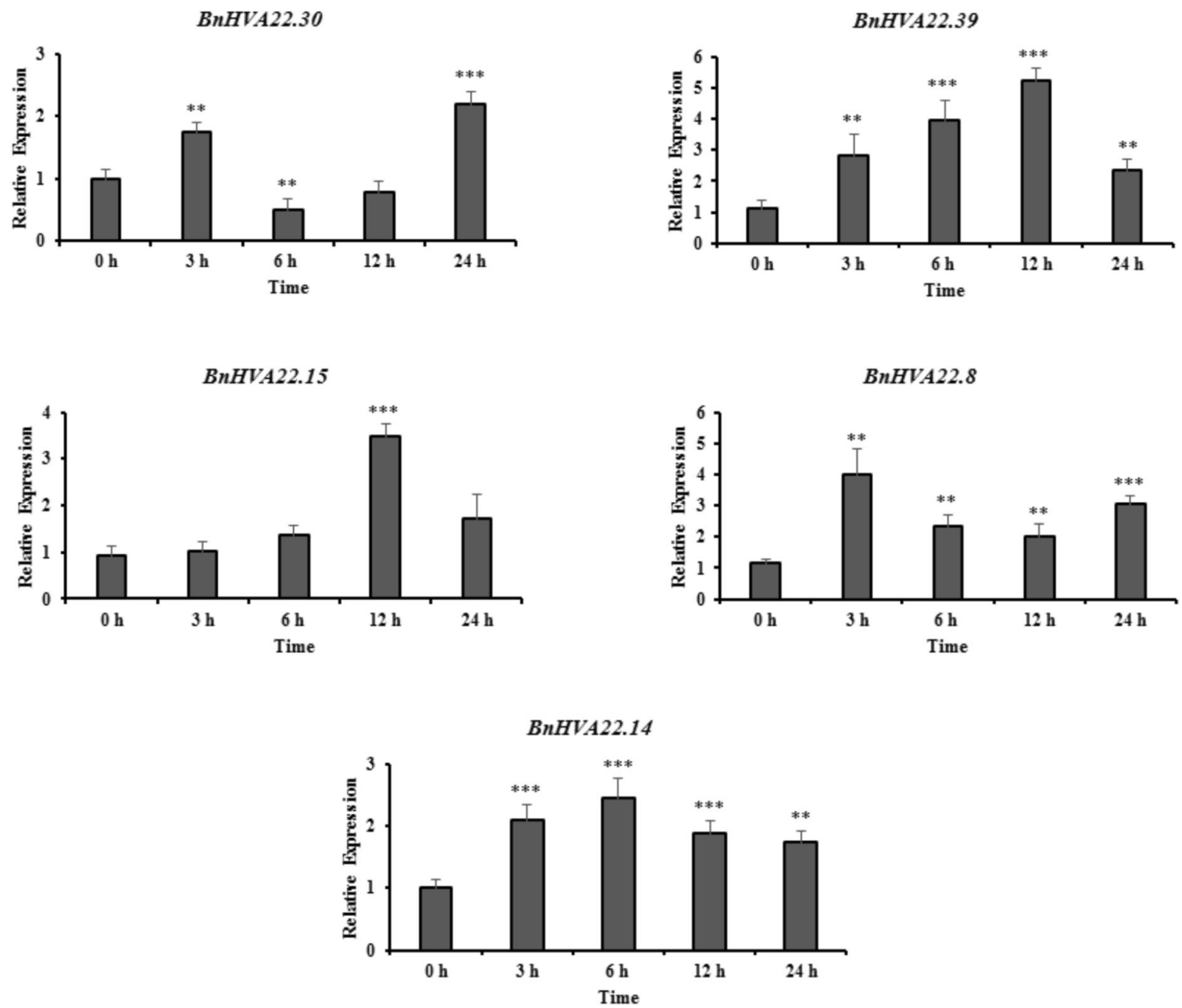


Fig. 9 qRT-PCR analysis of selected *BnHVA22* genes in response to ABA treatment (100 μM). Actin 7 was used as internal control. The asterisks indicate significant differences according to Student's t test (*$P < 0.05$, **$P < 0.01$, *** $P < 0.001$)

nucleotides. The conservation of intron phases decreases from phase 0 to phase 2 (Long and Deutsch 1999; Raboanatahiry et al. 2021). Our analysis further confirmed the conservation of intron phases within each *BnHVA22* subfamily. The D clade predominantly exhibits a 01110 intron phase pattern, while the C clade is characterized by the 0111 pattern. The A and B clades show 00121002 and 011100 patterns, respectively. Overall, 46.57% of introns are phase 0, 50.22% are phase 1, and 3.19% are phase 2, suggesting a high degree of conservation. These findings, combined with the analysis of conserved motifs and exon–intron structures, support the phylogenetic grouping of the *BnHVA22* gene family.

Duplication is a key driver of gene family expansion and speciation, providing the raw material for the development of new traits and enhancing plant adaptability to environmental stresses (Gao et al. 2019; Laha et al. 2020). Our study of gene duplication in the *BnHVA22* gene family demonstrated that segmental/WGD duplication is the primary mechanism responsible for the expansion of the *HVA22* gene family in *B. napus*. The role of segmental/WGD duplication in expanding the *HVA22* gene family has also been observed in wild tomato species, cotton species, and *G. max* (Chen et al. 2024; Zhao et al. 2023; Zhang et al. 2023a). Intraspecific collinearity analysis within the *B. napus* genome identified 27 pairs of *BnHVA22* paralogous genes, which are under purifying selection pressure, thereby preserving their functions during evolution (Abedi et al. 2021). Furthermore, interspecific collinearity analysis showed that 36 *BnHVA22* genes have orthologs in the genomes of *A. thaliana*, *B. rapa*, and *B. oleracea*, with 21 *BnHVA22* genes exhibiting collinearity relationships in all three species. It is important to note that *B. napus*, *B. rapa*, *B. oleracea*, and *A. thaliana* belong to the Brassicaceae family, thus exhibiting strong collinearity relationships among them.

Cis-regulatory elements serve as binding sites for transcription factors, initiating and regulating gene transcription. As such, the temporal and spatial regulation of gene expression is largely dependent on these crucial regulatory elements (Villao-Uzho et al. 2023; Kummari et al. 2020). Analysis of the promoter regions of *BnHVA22* genes revealed a diverse array of regulatory elements associated with various biological processes, including hormone signaling, stress responses, growth regulation, and developmental pathways. Research of *HVA22* gene promoters in other plants—such as *P. edulis*, *G. max*, *S. lycopersicum*, wild tomato species, cotton, and citrus—has similarly identified a variety of regulatory elements responsive to hormones and environmental stress (Ferreira et al. 2019; Wai et al. 2022; Zhang et al. 2023b; Zhao et al. 2023; Chen et al. 2024; Hou et al. 2024).

The most prevalent regulatory elements responsive to hormones in the *BnHVA22* genes are associated with ABRE and ERF, which play crucial roles in responses to ABA and ethylene, respectively. The *HVA22* genes are recognized as ABA-responsive genes (Shen et al. 1993). Both ethylene and ABA are essential for plant stress responses and developmental processes (Singh and Roychoudhury 2023) Therefore, it can be concluded that the *BnHVA22* genes contain *cis*-regulatory elements that respond to these hormones in pathways activated by ABA and ethylene (Chen et al. 2024). Additionally, the MYc and MYb regulatory elements, which are implicated in drought responses, were found to be the most frequently occurring among stress-responsive regulatory elements. Functional research of *HVA22* genes across various species has demonstrated that this gene is induced by cold, drought, and ABA, among other factors. Thus, the distribution of diverse *cis*-regulatory elements in the promoters of *BnHVA22* genes highlights their significant role in responding to environmental stress (Chen et al. 2002; Shen et al. 2001; Wai et al. 2022).

MicroRNAs (miRNAs) are short, non-coding RNA molecules (20–22 nucleotides) that control gene expression by interacting with mRNA, leading to its degradation or translation inhibition (Bajczyk et al. 2023). In our study, we identified miRNAs from the miR167, miR169, miR172, MiR6028, and miR6032 families as potential regulators of *BnHVA22* gene expression. Previous research has shown the involvement of these miRNAs in various biological processes. For example, miR167 enhances heat stress resistance in *Vitis vinifera*, increases resistance to clubroot disease in *B. rapa*, and influences cold stress response pathways in *Saccharum officinarum* (Liao et al. 2023; Song et al. 2024; Zhang et al. 2023b). Additionally, *miR169* negatively affects *Z. mays* resistance to *Bipolaris maydis*, while positively influencing the responses of *A. thaliana*, *Z. mays*, and *O. sativa* to abiotic stress (Xie et al. 2024; Chen et al. 2024).

To further understand the function of the *BnHVA22* genes, examining their expression profiles across different tissues and under various abiotic stress conditions is essential. Transcriptomic data revealed that *HVA22* genes in *B. napus* exhibit distinct expression patterns across tissues and in response to abiotic stress, emphasizing the diverse roles these genes play in growth, development, and stress responses. Based on these expression patterns, we classified the *BnHVA22* genes into three distinct groups. The first two groups consist of genes that are expressed in most tissues, including root, stem, leaf, silique, flower, and seed, as well as under conditions of salt, PEG, cold, heat, and ABA stress. In contrast, the third group of *BnHVA22* genes is tissue-specific and does not show significant expression under stress conditions. Differential expression of *HVA22* genes under various physiological conditions has also been reported in other plant species (Hou et al. 2024; Zhao et al. 2023; Wai et al. 2022; Zhang et al. 2023a). For instance, expression of the *HVA22* gene in *S. lycopersicum* shows an increasing trend during fruit growth and development (Wai et al. 2022). Additionally, in *G. hirsutum*, the *HVA22* gene is expressed in ovules and fibers, with the highest expression levels of *GhHVA22E* found in vegetative tissues (Zhang et al. 2023a).

Since *HVA22* genes are known to be responsive to stress, particularly ABA, we evaluated the expression patterns of several *BnHVA22* genes in response to salinity and ABA using qRT-PCR. The results revealed that, with the exception of *BnHVA22.30*, which did not exhibit a significant response to salt stress, all other genes showed notable changes in expression levels following salt and ABA treatments. These findings align with previous research indicating that *HVA22* genes are up-regulated in response to abiotic stress, especially salinity and ABA (Wai et al. 2022; Zhao et al. 2023; Hou et al. 2024; Zhang et al. 2023a; Chen et al. 2024). For instance, the expression of *SlHVA22j* in *S. lycopersicum* was significantly induced at all-time points following treatments with salt, drought, heat, and *ABA* (Wai et al. 2022). A similar expression pattern was observed for *GhHVA22E1D* in response to salt, drought, and *ABA* stress. Functional analysis of *GhHVA22E1D* revealed that its overexpression in *A. thaliana* enhanced the transgenic plant's tolerance to salt and drought, while silencing the gene in *G. hirsutum* increased the sensitivity of knockout lines to these stresses (Zhang et al. 2023a).

In this study, the expression patterns of *BnHVA22.8* and *BnHVA22.39* in response to salt and *ABA*, as well as *BnHVA22.16* in response to *ABA*, were found to closely resemble those of *SlHVA22j* and *GhHVA22E1D*. These findings suggest that these genes likely play comparable roles in mediating plant responses to abiotic stresses. Interestingly, some genes exhibited a biphasic expression pattern under salt and *ABA* stress, showing significant induction at certain time points and suppression at others. Specifically, *BnHVA22.14* and *BnHVA22.15* responded biphasically to salt stress, while *BnHVA22.30* showed a similar pattern under *ABA* treatment. A comparable biphasic response to salt and ABA stress has also been reported for *HVA22C* and *HVA22D* in *P. edulis* (Hou et al. 2024). This biphasic response may indicate complex regulatory mechanisms governing these genes.

This study highlights the potential application of the identified *HVA22* genes in improving stress tolerance in *B. napus* and other crops. The up-regulation of specific *HVA22* genes in response to stressors such as salinity and *ABA* underscores their role in abiotic stress resilience, making them promising candidates for breeding or genetic engineering efforts aimed at developing climate-resilient crops.

Conclusion

In this study, we identified 39 HVA22-encoding genes in the genome of *B. napus*, which were randomly distributed across the chromosomes of this species. Evolutionary analyses of the *BnHVA22* genes revealed that segmental/WGD duplication events were the primary drivers of the expansion of this gene family. Phylogenetically, these genes were grouped into four distinct clades, exhibiting conserved gene structures and motif patterns. The presence of stress- and hormone-responsive *cis*-regulatory elements, particularly those responsive to ABA, along with the regulatory effects of miRNAs on certain *BnHVA22* genes, highlights the complex transcriptional and post-transcriptional regulation of this gene family. Expression profiling based on transcriptomic data across tissues such as root, stem, leaf, flower, seed, and silique, as well as under stress conditions including salinity, PEG, cold, heat, and ABA treatment, alongside qRT-PCR analysis of selected genes under salinity and ABA stress, demonstrated that *BnHVA22.8*, *BnHVA22.14*, *BnHVA22.15*, *BnHVA22.30*, and *BnHVA22.39* are key members of the *BnHVA22* family. These genes likely play crucial roles in numerous biological processes in *B. napus*, particularly in responses to abiotic stresses. This study does not include functional validation of the identified *HVA22* genes, which could be achieved through gene knockout or overexpression experiments to clarify their precise roles in stress responses. Additionally, integrating multi-omics approaches could offer deeper insights into the molecular mechanisms underlying *HVA22* genes function in stress tolerance. Future research should address these aspects to validate the findings and further explore the regulatory networks governing *HVA22* genes activity.

Author contribution P.W.: Formal analysis, Methodology, Software, Writing—original draft; L.W.: Data curation, Conceptualization, Validation, Writing—Review and Editing.

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Data availability Data is provided within the manuscript or supplementary information files.

Declarations

Conflict of interest The authors declare no competing interests.

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